

IMP RTU TOPICS 3rd SEM

AEM Important topics

for 10 marks

unit 1 joint pdf with marginal distribution and conditional distribution

moment about mean questions

unit 2 finding rank of correlation coefficient with the help of spearman or carl pear son method and regression of line

unit 3 uses of optimization techniques and classification of Optimization tech.

unit 4 langrage multiplier method and KT condition

unit 5 Simplex method, two phase method, big M method and transportation-: VAM method with optimality test for 10 marks only (special case - assignment problem)

For 4 marks

unit 1 pdf and pmf questions , moment about mean and origin, skewness and kurtosis

Chebyshev inequality (2marks)**

unit 2 results of distribution functions like binomial , Poisson , exponential and normal mean and variance

unit 3 optimization problem formation

unit 4 langrage multiplier method , hession matrices method,

unit 5 transportation problem with North west corner method or least cost method, duality**,

DSA Imp Topics

Unit 1. and Unit 2

difference between stack and array, difference between queue and linked list, circular linked list, doubly linked list

tower of honoi, recursion, infix, prefix and postfix conversion, Notation(big O , Theta and omega)

Unit 3. sorting's

quick sort, merge sort ,radix sort, insertion sort, selection sort ,search, difference between binary and linear search

Unit 4. Types of tree : complete tree, balanced tree, B tree, B+ tree, BST, AVL tree and threaded tree ,In order pre order and post order traversal

Implementation of AVL, deletion in AVL tree, rotations LL LR RL RR

Unit 5 Dijkstra , Kruskal and prims algo, DFS and BFS in graph

OOPs Imp topics

Unit 1 programming paradigms, characteristics of OOP's, Class and structure difference, objects (in side and out side of the Class) , array of object (create data of employees in 10 marks)
Unit 2 Constructor and destructor, Pointer of object, function overloading
Unit 3 Inheritance (multiple, multilevel, simple, hybrid), virtual class and virtual function, abstract class (in 2 marks and 4 marks for short note)
Unit 4 Overloading, "+", "-" And "=" operator overloading, const and static keyword use, overriding
Unit 5 Class Template (Exp of real and imaginary number in 10 marks), exception (devide by zero and) and file handeling (enter number and save in even file and odd file in 10 marks)

MEFA Imp topics

Unit 1 Deductive and inductive analysis, Scope and nature of economics, managerial economics, National income , GDP,
Unit 2 Demand and supply function, demand and supply elasticity, demand forecasting,
Unit 3 Cost and production function, Different types of cost, long run and short run in cost
Unit 4 Comparison between all market (oligopoly, monopoly, monopolistic, perfect)
Unit 5 Solvency , liquidity and turn over ratio, Balance sheet, short definition of all financial analysis

DE Imp topics

Unit 1 Realization of other gates using universal gate, Code conversion, gray code, excess-3 code
Unit 2 Tabular method and k-map method, SOP , POS , min term and max term , don't care conditions
Unit 3 comparison between DTL, ECL, RTL,TTL,CMOS and MOSFET. characteristics of Logic family- power dissipation , fan-in, fan-out, operation Speed
Unit 4 Half and full adder, Implementation of Boolean expression using Mux, 2:1 and 4:1 mux ,
Unit 5 Shift registers , circuit diagrams of shift registers (10 marks), diff. between Latch and flip flop, excitation table and flip flops, Counters (concept of sync and async ,up down and mode counters)

SE Imp topics

Unit 1 Software engg and characteristics of good software, SDLC (mainly Increment, waterfall, Spiral and RAD model), diff between verification and validation, SRS documentation

Unit 2 Principle of Risk analysis, Requirement analysis, Cocomo model, FP and LOC (numericals in 10 marks), system quality assurance,

Unit 3 UML (use case diagram, DFD And CFD, FSM), Software prototyping, diff between Structural and behavioural modeling

Unit 4 Coupling and cohesion, Software design concept, SSD, architecture and procedural design diff

Unit 5 Diff between OOA and OOD, class and Object relationship and modularization

prepare short note on these topics so it can be easily extracted for 10 marks also